Algebra 1 Name_Answer Key_________
Unit 13 Date____________________
Lesson 10 Homework Period___________________

Given a written description, identify the function as linear, quadratic, or exponential. Explain your choice.

1. An investment grows at a constant percent rate of 2.25% APR.

   Exponential: constant percent will always be exponential.

2. The Occupational Safety and Health Administration (OSHA) allows for stairs to have a rise of 9 inches and a run of 10 inches.

   Linear: constant ratio of $9:10$.

3. A ball is thrown from a height of 5 feet. Four seconds later, it reaches a maximum height of 25 feet.

   Quadratic: follows a quadratic curve it will go up then fall back down with change in direction.

4. A rare species of fish experiences a population decline at a constant percent rate of 4.5%.

   Exponential: constant percent will always be exponential.

Given a table of values, identify the function as linear, quadratic, or exponential. Explain your choice.

5.

x	y
0	120
1	111
2	102
3	93
4	84

Linear, constant 1st difference

6.

x	y
0	165
1	179
2	195
3	211
4	230

Exponential, no constant 1st or 2nd difference but ratio ≈ 0.92.

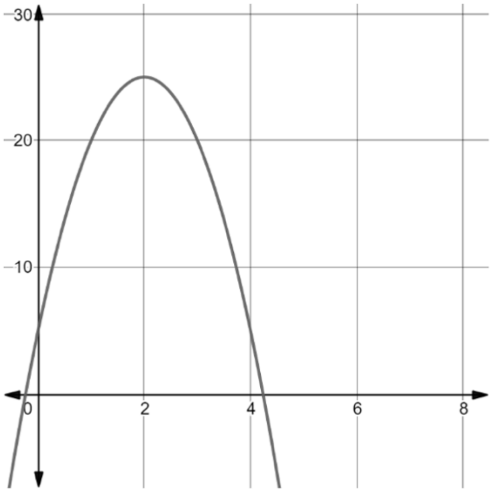
7.

x	y
1	20
2	35
3	40
4	35
5	20

Quadratic, constant 2nd difference

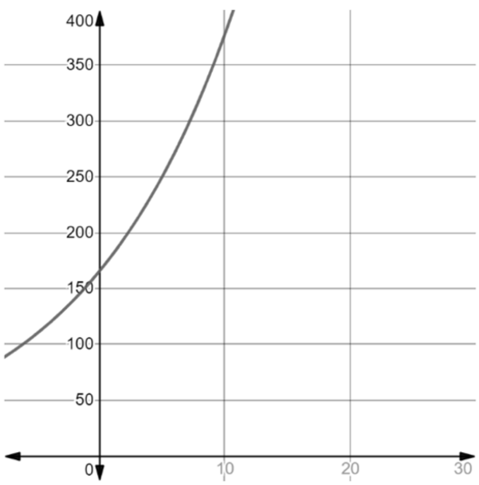
Given a graph, identify the function as linear, quadratic, or exponential. Explain your choice.

8.



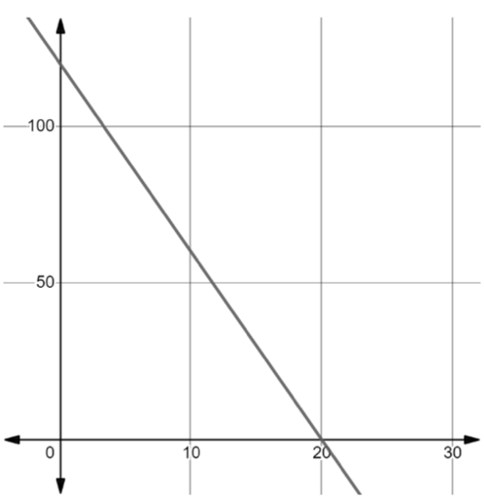
Quadratic
-follows quadratic curve
-constant second difference in y-values

9.



Exponential
-constant ratio from one y-value to the next
-follows exponential curve

10.



Linear
-straight line
-constant slope